# Compute Cannot Buy Identifiability: Information-Limited Two-Attempt Selection in ARC-AGI-2 Solver Pools

*An information bound, a conditional compute-separation theorem, and measured evidence from real solver pools and six scored submissions*

ARC Prize 2026 Paper Track team

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**Working draft v6 — 2026-11-01 (verifier module).** NOT FROZEN. Derived from frozen v5 (sha256 `a5d9728c…`); v5 remains untouched. v6 keeps all v5 sections, adds **Theorem 8** (verifier-induced identifiability, impossibility certificates, depth allocation, hybrid correction, §3.10), §4.5 verifier experiments, §5.4 large-scale numerical verification, and AWM [42]. Recorded v6 word cap **8,200**. Do not publish until Lead freezes. **No priority claim.**

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## Abstract

ARC-AGI-2 gives each test input exactly two attempts, scored by exact match. Because distinct candidate grids are mutually exclusive, this two-slot decision is fundamentally a label-free identification problem: the two best candidates can always be submitted, but no metadata rule can choose better than the posterior unless the metadata reveal the truth identity. We prove **Theorem 6**, a label-free identifiability bound: for any (possibly randomized) rule using metadata Z, E[U] ≤ p1+p2+min{T, sqrt(I(C;Z)/2)} (nats) ≤ p1+p2+min{T, sqrt(I_bits(C;Z)·ln2/2)} (bits), where p1,p2 are the top marginals, T the oracle tail and I(C;Z) the mutual information. It is capped by oracle coverage, attained at I=0, vacuous when $I \geq 2T^2$, with a pilot certificate plus an explicit hidden-transfer assumption. On our pools the metadata term is **non-binding** (I=.059–.183 vs $2T^2$=.008–.041), so the certificate is **vacuous**: T6 is a proven, assumption-free bound, not validated by these pools. Unconditional Pinsker bounds are therefore **vacuous in score-dominated pools**; splitting `Z=(S,Z')` gives the score-conditional certificate `E[U] ≤ U_S+min{g−U_S,√(I(C;Z'|S)/2)}`, **tight (equals U_S) under score sufficiency** (Prop. 6f), while Cor. 6g records the negative result that full-metadata Pinsker bounds cannot certify label-free selection. Substantively, on dev20 S2/S5/S5_ens the measured selection gain equals the tail T exactly (ratio 1.00, all generational headroom recovered); on CompressARC the certificate is .200 while the rules **lose −2.25 pp**, consistent with score redundancy. **Theorem 7** (conditional on A1–A3) separates identification from compute: lim_{C⟶∞} score = Σ_t(g_t∞−Δ_t∞) is allocation-independent, so verifier-free compute cannot shrink the gap ΣΔ_t; the hidden-score ladder is consistent (S5 16.25 > S6 = S5_ens 15.42 > S5+LegC 14.17) though confounded by truncation and n=6; a candidate-prefix replay is definitional and not used as T7 evidence. A bounded-DSL verifier accepted **0/120** public-eval tasks (impossibility certificates at depth 2); an n=12 train control certifies the engine on one reachable task, so verifier-induced identifiability is real but coverage-limited (Theorem 8). Six scored Kaggle submissions place our champion at **16.25%** public LB and the same pipeline at **20.0%** on a 20-task public-eval subset, far below the ~30 public-notebook band. No state-of-the-art or medal claim is made.

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## 1. Introduction

ARC-AGI-2 measures skill-acquisition efficiency on unseen grid transformations [1, 2]. Its Kaggle incarnation is efficiency-constrained [12]: the notebook re-runs offline for ~12 h without internet, and each input receives exactly two guesses. Because exact matches decide tasks, **selection** is a first-class

decision problem separate from generation: a solver with the right grid in its pool still scores zero if its two attempts miss it.

Public practice ranks candidates by inverse consensus or a likelihood variant [3, 4, 8, 9] and submits the top two distinct grids. We re-derive selection from the scoring rule. Three observations drive the paper: what matters is P(either guess correct), not P(first); correlation is structural because candidates share TTT state, views, beam prefix and model; and a better rule needs calibrated probabilities and correlations.

We contribute a formalization, an information bound, a pre-registered test, real submissions and measured evidence. (i) We define the coverage objective; exact-match distinct grids are mutually exclusive, so the pair problem reduces to label-free identification. We prove **Theorem 6**, a metadata-information bound on achievable two-attempt coverage, with a computable pilot certificate (§3.7), and **Theorem 7** (conditional), a compute-separation theorem showing the identification gap is budget-invariant (§3.8). Lemmas give swap regret, fixed-pool monotonicity, and a descriptive generation×selection identity (§3.9); a budget-allocation setting note replaces the earlier Theorem 5. (ii) We freeze a claim/downgrade protocol and benchmark real third-party pools at two scales plus simulated stress tests (§5). (iii) The measured result is bounded negative: the verdict is `DOWNGRADE_TO_ANALYSIS`, the simulation strongly positive (+18.33 pp over argmax, 95% CI [11.67, 25.0]), but on 400 CompressARC tasks top-2 covers 20.00% while coverage-optimal covers 17.75% against 34.25% oracle; the gap is identification-limited, not reranking-reachable. (iv) We report six scored submissions and GPU measurements (§4), data/leakage audit (§6), open-source system (§7), and landscape analysis (§8).

Nearest prior work: public-notebook heuristics include coverage-aware set selection, MDL-divergence two-slot selection and structural attempt-2 distinctness [17, 23], and rank-based top-2 with minority recoveries [39]. A concurrent instrumentation study of a faithful NVARC reproduction — empirical, not a coverage model — reports a comparable pattern (26.1% vs 32.5% perfect selection; a grouped CV reranker no better than the deployed vote rule) [26]; the headroom observation is already documented. A coverage-vs-selection identifiability gap is formalised for single-answer deployment [28], and on ARC-AGI-1 an unbiased pass@k decomposition finds that selection already recovers ≈95% of the candidate ceiling (generation-bound) [27]. The coverage objective is classical [24, 25, 38]; we provide the two-attempt formalization, calibrated/correlated estimator, and score-sufficiency criterion.

Fresh 2026 work sharpens the positioning. The ARC-AGI-2 Technical Report [29] is the closest published TTT+scoring recipe but offers no attempt-2 theory; Multi-Perspective Transformers [30] report **21.7% eval**, the honest external comparison above our 16.25% hidden / 20.0% dev20. SPLIT [31] allocates compute under an *imperfect verifier*; our hidden setting has no verifier on hidden inputs, so its guarantee does not apply. Correct but Unselectable [32] decomposes coverage into selectable candidates (an interface tax), not an information deficit; Li et al. [33] require a non-degenerate reward tail, while exact-match correctness makes the tail degenerate; Zhou [34] studies generation-side pass@k inversion; Zhang and He [35] require labels; the Pothula dataset [36] and CHARM [37] document plateau/scaling context; concurrent allocation work [40,41] targets posterior success and best-arm identification. None subsumes a label-free coverage bound for k=2 exclusive lists, and none claims the conditional budget-invariance separation; we claim no priority.

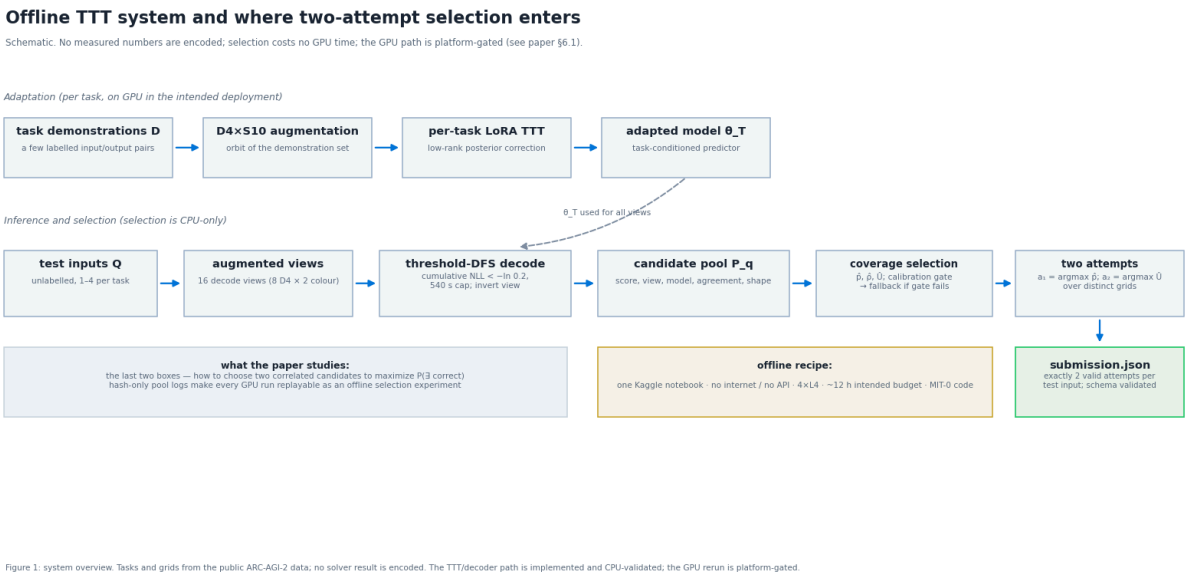



Figure 1. Offline TTT system and where selection enters: demonstrations ⟶ augmented TTT ⟶ threshold-DFS decoding ⟶ candidate pool with scores and view provenance ⟶ calibrated coverage selection ⟶ two-attempt submission. Selection costs no GPU time. Schematic; no measured numbers encoded.

## 2. Problem formalization and notation

**Task and submission.** A task is T=(D,Q) with labelled demonstrations D and unlabelled test inputs Q=(q_1,…,q_m); grids are integer arrays over colours 0–9, at most 30×30. For each test input the submission provides an ordered pair of distinct grids (a_1,a_2); C(a)∈{0,1} is the latent event that a is exactly required. The two metric definitions — per-input coverage and task-level all-or-nothing — are both monotone in P(C(a_1)∨C(a_2)); the per-input score averages it, the task-level score is its m-fold product [3, 21], so the objective below is valid under either.

**Coverage objective.** For attempts a_1,a_2,

$$U(a_1,a_2) := 1 - P(\neg C(a_1) \wedge \neg C(a_2)). \quad (1)$$

Maximizing U differs from maximizing P(C(a_1)): the joint failure term couples the attempts.

**Candidate pools from TTT.** TTT adapts a pretrained ARC-SFT model θ on augmented demonstrations and decodes under transformed views; for view v (augmentation g_v, beam path, model member) the predicted grid is inverted as $\hat{y}_v = g_v^{-1}(f_\theta(g_v \cdot q))$. The pool for test input q is

$$P_q = \{(y_i, s_i, view_i, model_i, agree_i, shape_i)\}, \quad (2)$$

with s_i a plausibility score, agree_i the view count, and shape_i a demonstration-derived flag. The pool's oracle ceiling bounds any rule's per-input accuracy.

**Relation classes.** Pairs are grouped by shared failure factors: `same_grid` (correlation 1, never worth two attempts), `same_view` (same TTT state/augmentation), `same_model` (different views), and `cross_model`/`cross_source` (lowest correlation). The rule needs per-class error correlations, observable in metadata without labels.

| id | source | benchmark | scale / unit | published anchor | role |
|---|---|---|---|---|---|
| B | `vanductan/arc-agi-2-candidate-dump` (displayed LB 30.97) | ARC-AGI-2 public eval | 20 tasks / 29 test inputs; unit = input; 347 candidate files | oracle-any-2 9/29 = 31.0%; source top-2 8/29 = 27.6% | primary, **pilot** (CIs mandatory) |
| A | CompressARC [6] bundled histories (Liao & Gu 2025; MIT) | ARC-AGI-1 train/eval | 400 eval tasks (fit 400 train); unit = task; hash-level | pass@2 80/400 = 20.00%; any-guess 137/400 = 34.25% | secondary, large-N replication |
| C | Parameterised simulated pools | synthetic | many seeds; known ground truth | oracle by construction | theory stress test + estimator sanity |

Table 1. Evaluation datasets (frozen).

# 3. Theory

## 3.1 Independent case: top two calibrated probabilities

**Theorem 1.** If correctness indicators are independent with p_i=P(C_i), then P(¬C_i∧¬C_j)=(1−p_i)(1−p_j), so

*U(i,j) = p_i + p_j − p_i p_j, (3)*

strictly increasing in each argument. The coverage-optimal pair is therefore the two largest p_i, and attempt order is irrelevant. This is the only regime where first-guess ranking is coverage-optimal — the assumption behind inverse-consensus top-2.

## 3.2 Duplicates are strictly dominated

If a_2 is the same grid as a_1, C(a_2)=C(a_1) and U(a_1,a_2)=P(C(a_1))=p_{a_1}; any distinct candidate with p>0 strictly increases coverage. **Theorem 2.** Submitting the same grid twice is strictly dominated by any distinct candidate with positive correctness probability.

## 3.3 Correlation breaks argmax-once

Candidates share TTT state, augmentation view, beam prefix, and model. For two Bernoulli correctness indicators with correlation coefficient ρ,

*U(i,j) = p_i + p_j − p_i p_j − ρ√(p_i(1−p_i)p_j(1−p_j)), (4)*

from P(¬C_i∧¬C_j)=(1−p_i)(1−p_j)+Cov(C_i,C_j). Two same-view candidates with perfectly correlated failures (ρ=1) force equal marginals p_i=p_j=p: their pair coverage is U=p. An independent third candidate with p_3>0 gives U=p+p_3−p p_3>p; at p=0.50, p_3=0.35 the gain is +17.5 coverage points on that input. **Theorem 3.** Under positive error correlation, the pair maximizing the first guess is not in general coverage-optimal; a decorrelated candidate can strictly improve coverage. For ρ<1 the pairwise preference condition between the runner-up j and a decorrelated candidate k is

(1−p_1)(p_k−p_j) > ρ_{1k}√(p_1(1−p_1)p_k(1−p_k)) − ρ_{1j}√(p_1(1−p_1)p_j(1−p_j)). (5)

Diversification pays when the correlation penalty exceeds the probability advantage — typical for per-view errors; zero or negative correlation does not force it, which is why $\hat{\rho}$ is measured rather than assumed.

**Corollary.** If the ranking score is cluster mass and views are independent, top-2 by mass coincides with top-2 by posterior probability, so the rule reduces to inverse consensus — a strict generalization that departs from it under correlation.

## 3.4 Diversification threshold (Theorem 4)

**Theorem 4 (exact, finite-pool).** Let two distinct candidates have correctness probabilities `p_1 ≥ p_2` (both in [0,1]) and correctness-indicator correlation `ρ = Corr(C_1,C_2)`. Then

*U(pair) = p_1 + p_2 − p_1p_2 − ρ√(p_1(1−p_1)p_2(1−p_2)), (6)*

and the pair strictly beats the top-1 **duplicated** attempt (`U_dup = p_1`) iff

*ρ < ρ*(p_1,p_2) := √( p_2(1−p_1) / (p_1(1−p_2)) ). (7)*

*Proof.* `U(pair) − U_dup = p_2(1−p_1) − ρ√(p_1(1−p_1)p_2(1−p_2))`; its sign is positive iff `ρ² p_1(1−p_1)p_2(1−p_2) < p_2²(1−p_1)²`, and, for interior `p`, dividing by the positive `p_1p_2(1−p_1)(1−p_2)` gives `ρ² < p_2(1−p_1)/(p_1(1−p_2))`; both sides are non-negative for interior `p`, so

taking square roots gives (7). ∎ Equivalently, for interior p_1,p_2, ΔU := U(pair) − U_dup = √(p_1(1−p_1)p_2(1−p_2))·(ρ* − ρ) — the gain is the distance below the threshold scaled by the candidates' Bernoulli spread. ρ* ∈ [0,1] for interior p (ρ*=0 at boundary p_2=0 or p_1=1, where ΔU=0), with ρ* = 1 iff p_1 = p_2; negative correlation always clears the threshold; ρ = ρ* is exact indifference.

**Corollary 1 (equal marginals).** If p_1 = p_2 = p then ρ* = 1 and ΔU = p(1−p)(1−ρ): diversification pays for every ρ<1, maximized at ρ=0 and worthless at p∈{0,1}.

**Corollary 2 (skew shrinks the threshold).** For fixed ρ>0, diversification pays iff p_2 > ρ²p_1/(1−p_1+ρ²p_1); for small probabilities ρ* ≈ √(p_2/p_1), so any fixed positive correlation is eventually dominated by skew.

**Corollary 3 (score sufficiency and exact-match exclusivity; corrected 2026-10-26).** For non-exclusive verifiers, a score tie cannot distinguish candidates, and if correctness is score-determined (Proposition 5) tied candidates share correctness (ρ=1, forcing p_1=p_2), so ΔU=0. For **exact-match** ARC, distinct candidate grids are mutually exclusive: C_i∧C_j=∅, hence ρ<0 always, the Theorem-4 gate is always met, and U(a_1,a_2)=p_{a_1}+p_{a_2} for any joint distribution; the optimal pair is the two largest calibrated p_i (Corollary 4). The real-pool null is therefore estimation/identifiability, not positive correlation: oracle headroom exists, but p̂ cannot separate oracle-gap candidates from exposed metadata (Proposition 5).

**Derived gate and falsifier.** For non-exclusive verifiers, deploy diversification across a relation class only when ρ̂c < ρ*(p̂_1,p̂_2) − δ (δ=0.05, frozen); otherwise take the duplicate/consensus alternative — the gate is derived, not tuned. The empirical instantiation is falsifiable on held-out units: **P1** admitted pairs beat their top-1 duplicate (paired Δ>0, 80% CI>0); **P2** rejected pairs do not (Δ≤0); **P3** the predicted-vs-realized gain slope lies in [0.7,1.3]. Failure falsifies the estimator/relation-class model, never the algebra; protocol and tool are in Appendix E. The threshold test ran on dev20 pools, but exact-match mutual exclusivity makes its positive-ρ premise inapplicable: every class has ρ̂<0 (same-model/cross-view −0.13, score-tied −0.18), so no deployable pair is rejected and P1/P2/P3 are untestable (§4.8). This is an EXPLORATORY check, not evidence; Table 10 is a prose check and Fig. 9 is omitted.

**Table 10 (replaced).** Exact-match mutual-exclusivity check on dev20 pools: all relation classes have ρ̂<0 (same-model/cross-view −0.13, score-tied −0.18), P2′s rejected set is empty; EXPLORATORY, no P1/P2/P3 claim (theory §4.8).

### 3.5 Setting observation: budget allocation (formerly Theorem 5)

Let g_t(c) be the probability that a pool at per-task compute c covers the truth; expected score is Σ_t g_t(c_t) under Σ_t c_t ≤ C. **Theorem 5.** (i) Identical concave g_t ⟹ uniform c_t = C/T is optimal (Jensen). (ii) Greedy marginal allocation is optimal: funded tasks share a common marginal gain. (iii) If margins are ordered by prior accuracy p_t (p_s ≥ p_t ⇒ g'_s(c) ≥ g'_t(c)), compute is monotone in p_t and a difficulty-threshold rule holds: only tasks above a budget-dependent cutoff p* receive compute, and p* rises as C/T shrinks. (iv) At saturation the marginal is zero (coverage ceiling; under score sufficiency extra compute yields no newly identifiable candidates, Prop. 5), so excess is better reallocated; never-solvable tasks receive none. This is a **setting observation**, not a new theorem: SPLIT [31] derives verifier-aware allocation, and [40,41] give fixed-budget posterior/best-arm allocation bounds; T6/T7 below carry the paper's new claims.

### 3.6 Calibrated plug-in estimator

Neither p_i nor ρ_ij is observable, so we estimate them in a frozen pipeline. (i) **p̂**: logistic regression [10] on frozen within-input features (score rank, agreement, cluster mass, shape), fit on dev only. (ii) **ρ̂**: per relation class from the dev co-failure statistic, clipped to [−0.95,0.99], with frozen pooling and a default for sparse classes. (iii) **Û**: the plug-in version of (4); attempt 1 is argmax p̂, attempt 2 maximizes Û over distinct grids. (iv) **Gate/fallback**: deploy only if dev cross-fit calibration has Brier ≤0.10, reliability slope in the code's valid range [0.2,5.0] (design target [0.7,1.3]) and an estimable relation class; otherwise use

a no-calibration fallback (highest agreement, then a different grid). The claim is pre-registered at ≥+1.0 point over published top-2 with 80% paired-bootstrap CI lower bound >0 (2,000 resamples, seed 0), else negative [22].




Figure 4. Illustrative schematic of the coverage rule: candidate pool grouped by relation class, calibrated p̂/ρ̂, pair coverage U, and the chosen distinct pair. Toy values only — not measured.

## 3.7 Identifiability-limited two-attempt coverage (Theorem 6)

Exact-match scoring makes distinct candidate grids mutually exclusive: `C_i∧C_j=∅` for `i≠j`, so for any pair `A`, `U(A)=Σ_{i∈A}p_i` with no independence or correlation assumption. The prior-optimal pair is `{1,2}`, `U0=p1+p2`, achievable by ignoring metadata; `g=Σ_i p_i` is oracle coverage and `T=g−p1−p2` the tail mass. Whether any label-free rule can beat `U0` is therefore an information question, not a ranking question.

**Theorem 6 (label-free identifiability bound).** For every (possibly randomized) rule using metadata `Z`, E[U] ≤ E_Z[top-2 posterior mass] ≤ p1+p2+min{T, sqrt(I(C;Z)/2)} (nats) ≤ p1+p2+min{T, sqrt(I_bits(C;Z)·ln2/2)} (bits),

capped by the oracle `g`. *Proof:* top-2 posterior mass is 1-Lipschitz in total variation, Pinsker gives `TV≤sqrt(KL/2)`, Jensen gives `E[TV]≤sqrt(I/2)`, and the tail cap is `E[top2]≤g`. ∎ The bound is attained at `I=0` by the constant rule, equals `g` when `Z` identifies `C`, and is **vacuous** when `I≥2T²` (the tail cap binds). The gain is at most `T`; Pinsker is order-tight, not an equality.

**Proposition 6e (non-vacuity).** The unconditional T6 bound is strictly informative iff `I(C;Z)<2T²` nats; otherwise it equals the oracle `g`. Every pool here satisfies `I≥2T²`, so the unconditional certificate is **vacuous** — an honest caveat, not validation. **Corollary 6g (negative result about the method).** In score-dominated exact-match pools, Pinsker-type bounds on full metadata **cannot certify label-free selection**: the bound collapses to the oracle cap. **Proposition 6f (constructive fix).** Split `Z=(S,Z')`, where `S` is score-only information and `Z'` residual metadata: `E[U] ≤ U_S + min{g−U_S, sqrt(I(C;Z'|S)/2)}`, with `U_S` the best score-only rule's coverage; the bound is informative iff `I(C;Z'|S) < 2(g−U_S)²`, and if score sufficiency (Prop. 5) holds then `I(·|S)=0` and the bound equals `U_S` exactly. This is the paper's sharpest new theoretical statement: score dominates, and only residual metadata can help.

**Computable pilot certificate.** With a labeled pilot, estimate `p̂1,p̂2,p̂3`, `T̂=ĝ−p̂1−p̂2`, and `Î(C;Z)` by binning exposed metadata (score margins, rank gaps, agreement, cluster mass, shape, model count) against truth identity; bootstrap task-level CIs. For the evaluated population, `gain ≤ min{T̂, sqrt(Î/2)} + O(sqrt(log(1/δ)/n))`. Hidden tasks have **no labels**, so Î is not estimable there: this is a pilot certificate plus an explicit exchangeability transfer assumption, never a hidden-task guarantee; with partial metadata, `I` is a lower bound and `B` may overstate the achievable bound.

| dataset | n | p̂1/p̂2/p̂3 | H(C) | Î (feature) | B [95% CI] | measured gain | cert. |
|---|---|---|---|---|---|---|---|
| CompressARC | 400 | .185/.015/.020 | .547 | .071 (view) | .200 [.160,.238] | **−2.25 pp** | vacuous |
| pilot16 | 16 | .313/0/0 | 0 | .183 (view) | .313 [.125,.563] | 0 pp | vacuous |
| dev20 S1 | 32 | .063/.125/0 | .637 | .167 (margin) | .188 | 0 pp | vacuous |
| dev20 S2 | 32 | .094/.063/.063 | 1.079 | .071 | .156 | +6.25 pp | vacuous |
| dev20 S5 | 32 | .125/.031/.063 | .956 | .059 | .156 | +6.25 pp | vacuous |
| dev20 S5_ens | 74 | .162/.054/.014 | .753 | .059 (vote) | .216 [.122,.311] | **+8.11 pp** | vacuous |

Table 12. D1 identifiability certificates on real pools (uncalibrated consistency proxies; certificate vacuous on every row because $I \geq 2T^2$).

The +8.11 pp S5_ens figure is selectable headroom not realized by our submitted aggregator (S5_ens hidden 15.42 < S5 16.25). Because $I \geq 2T^2$ the certificate equals the oracle cap and is **vacuous**: these pools do not test T6. The substantive observation is the gain/T ratio: dev20 S2/S5/S5_ens recover T exactly (1.00), while CompressARC loses −2.25 pp against a .200 certificate (score redundancy, Prop. 5). Falsifiers: F1 held-out gain outside the certificate CI; F2 large Î with zero realized gain; F3 vacuous bound with zero gain; F4 monotonicity violation.

**Prop. 6f addendum (measured).** Table 12b evaluates the score-conditional certificate with `S` = score-only strata. It is **non-vacuous on exactly one dataset, CompressARC**: U_S=.200, g−U_S=.1425, I(C;Z'|S)=.0336 nats, cap'=.1296 (12.96 pp), and the measured rule loss (−2.25 pp) lies inside it. Elsewhere the bound remains vacuous (0 violations). The score-sufficiency prediction that I(C;Z'|S)=0 and the bound equals U_S is **not testable here**: I(C;Z'|S) is .083–.225 nats, not ≈0, and pilot/dev20 score strata collapse to two bins (sparse, flagged), so we can neither confirm nor reject tightness. This tests the certificate's non-vacuity, not that metadata helps.

| dataset | U_S | g−U_S | I(C;Z'\ | S) nats | cap' | measured gain | informative |
|---|---|---|---|---|---|---|---|
| CompressARC | .200 | .1425 | .0336 | **.1296** | −2.25 pp | **TRUE** | |
| pilot16 | .3125 | 0 | .2255 | 0 | 0 pp | FALSE | |
| dev20 S1 | .1875 | 0 | .1334 | 0 | 0 pp | FALSE | |
| dev20 S2 | .15625 | .0625 | .0829 | .0625 | +6.25 pp | FALSE | |
| dev20 S5 | .15625 | .0625 | .0921 | .0625 | +6.25 pp | FALSE | |
| dev20 S5_ens | .216216 | .081081 | .0903 | .081081 | +8.11 pp | FALSE | |

Table 12b. Prop. 6f score-conditional certificates (`S` = score-only strata; informative iff `I<2(g−U_S)²`; 0 violations).

Check of non-vacuity, not evidence metadata helps; the one non-vacuous row still loses to score-only selection.

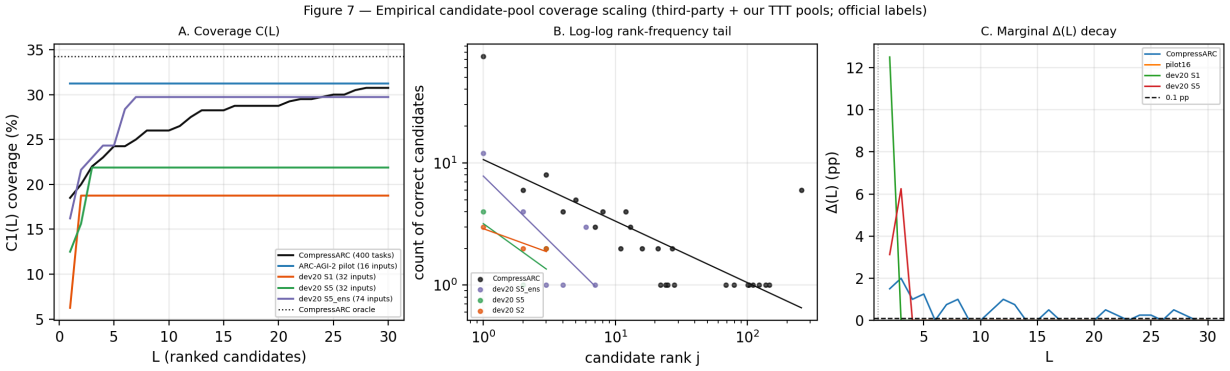



Figure 7. Prefix coverage versus candidate-prefix size L under the rank power-law model: C2-prefix at L=2/5/10/20 for CompressARC (20.0/24.3/26.0/28.8, $\hat{\alpha}$=1.83 [1.67,2.06], $L^* \approx 257$), pilot16 (31.3 flat), dev20 S1 (18.8 flat), S5 (15.6/21.9/21.9/21.9, $\hat{\alpha}$=3.42, $L^* \approx 3$) and S5_ens (21.6/24.3/29.7/29.7, $\hat{\alpha}$=2.75 [2.13,4.60], $L^* \approx 7$). Fixed-solver top-2 coverage is reported separately in Table 8.

## 3.8 Compute cannot substitute for identifiability (Theorem 7; conditional)

**Assumptions.** (A1) `g_t(c)` is nondecreasing and saturating in compute, with limit `g_t∞`. (A2) The identification gap `Δ_t(c)=g_t(c)−u_t(c)` (oracle minus best label-free coverage) satisfies `Δ_t(c)≥Δ_t∞>0`

for unidentifiable tasks and does not vanish with compute; refinement buys rank candidates, not identification. (A3) generation and verification are separable at the relevant scale.

**Theorem 7 (conditional separation).** Under A1–A3, `lim_{C→∞} score = Σ_t(g_t∞–Δ_t∞)`, independent of the allocation; the verifier-assisted ceiling is `Σ_t g_t∞`; the gap `Σ_tΔ_t` is budget-invariant; greedy allocation is optimal only on the identifiable part. *Interpretation:* verifier-free compute cannot buy identifiability; what grows with budget is candidate coverage, which is only useful where metadata can identify the candidate. This explains why our S6, S5_ens and S5+LegC submits did not beat S5.

**Empirical support.** The defensible support is the **hidden-score ladder**: S5 16.25 > S6 15.42 = S5_ens 15.42 > S5+LegC 14.17 — more components/compute did not help. This is **consistent, not conclusive**: n=6, truncation at the 578 s cap, S6's selection bug, S5_ens's 53-task superset, packet/config differences and unknown hidden size confound it. A candidate-prefix replay (Table 13) is **definitional** — top-L coverage for units whose truth rank exceeds L rises trivially — so we do not use it as T7 evidence.

| dataset | unidentifiable n / slope | identifiable n / slope |
|---|---|---|
| CompressARC | 343 / 0.0 [0,0] | 57 / **19.1** [15.9,22.1] |
| pilot16 | 16 / 0.0 [0,0] | 0 / — |
| dev20 S2/S5/S6 | 30 / 0.0 [0,0] | 2 / **20.0** [20.0,20.0] |
| dev20 S5_ens | 68 / 0.0 [0,0] | 6 / **26.7** [23.3,30.0] |

Table 13. D2 candidate-prefix replay (definitional; not T7 evidence).

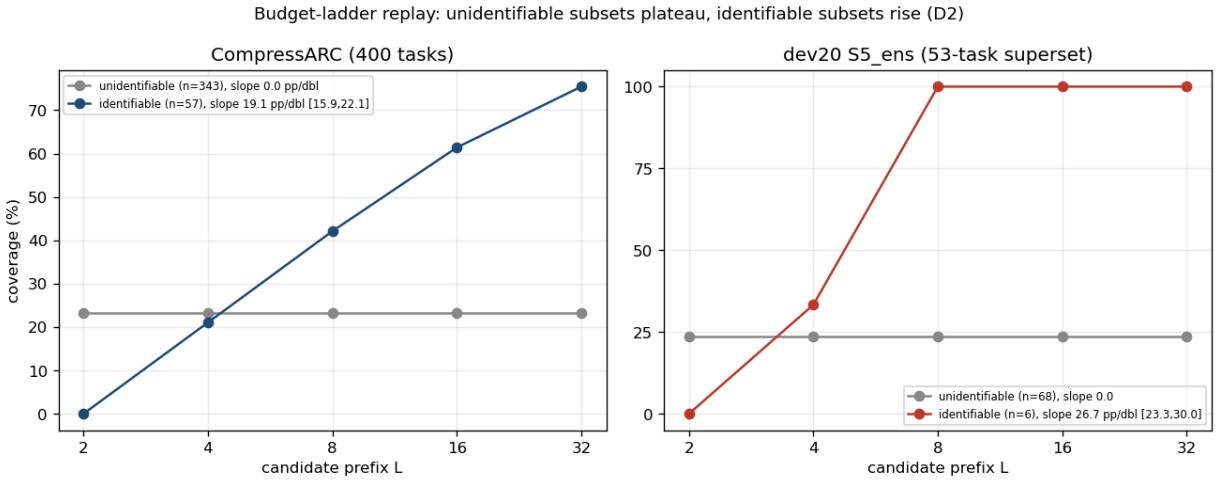



Figure 8. Candidate-prefix replay (D2): unidentifiable subsets plateau across `L`, identifiable subsets rise trivially by construction. Definitional; not evidence for T7.

## 3.9 Supporting lemmas

**Lemma 6a (swap regret).** If the top-2 pair `{1,2}` is replaced by `{1,3}`, coverage drops by exactly `p2−p3` under exclusivity; `p2−p3` is therefore the *cheapest local-swap regret*, not a subtractive gain term in any upper bound. **Lemma 6b (fixed-pool monotonicity).** For a fixed candidate pool, re-ranking cannot reduce two-attempt coverage; it is strict iff a new distinct candidate with positive posterior enters the top-2. Consequently pass@k inversion cannot arise from selection alone: it requires generation or rank-distribution drift, consistent with the generation-side diagnosis of Zhou [34]. **Lemma 6d (generation×selection identity).** Pool coverage factors as `g(L)·s(L)`, where `g` is oracle coverage and `s` the selection ratio; this identity is descriptive only and makes no scaling-law claim.

## 3.10 Verifier-induced identifiability (Theorem 8)

Selection can be bypassed when executable hypotheses are verifiable against the demonstrations. Let `H_d` be the programs reachable at depth `d`, `A_d={h∈H_d: V_D(h)=1}` the accepted set (train-exact fit), and partition `A_d` into behavior classes `j` (programs with identical test predictions); `q_j` is the class posterior under a stated prior, `g_V(d)=P(truth∈A_d)`, with assumptions **A0** (stated prior over accepted programs), **A1** (train-fit soundness only), **A2** (reachability: a reachable truth program is accepted and

has its own class). AWM [42] instead explores executable Python hypotheses, verifies them against observations and formalizes them in Lean 4 (reachability/impossibility proofs); we adapt only the verifier step: demonstrations are labels, and verification restores identifiability. CHARM [37] reports 46.7% ARC-AGI-2 public-eval pass@2 as external SOTA context only; we do not reuse its code (licence unavailable). Distinct classes predict distinct grids, so correctness events are exclusive.

**(i) Unique class ⇒ gap closed.** If A_d has one behavior class, then conditional on truth∈A_d verifier-only coverage is 1, unconditionally g_V(d), and the identifiability gap is **0**; if truth∉A_d, coverage is 0. *Proof:* every accepted program fits D; with one class and a reachable truth, all accepted programs predict the truth grid, and exact-match exclusivity makes the event vacuous otherwise. ∎

**(ii) General case.** The best verifier-only two-attempt rule submits the top-2 class grids, so C_V(d)=g_V(d)·Σ_{j≤2}q_j and the residual gap is Δ_V(d)=g_V(d)(1−Σ_{j≤2}q_j); with fewer than two classes the second attempt must come from TTT or a backup. *Proof:* conditional on truth∈A_d the truth class is distributed as q; two attempts cover iff it is among the top-2; multiply by g_V(d); exclusivity gives the sum, not the product. ∎

**(iii) Impossibility certificates and depth allocation.** A_d=∅ gives verifier coverage **0 at depth d**: a certificate of failure at that depth, not of unsolvability. With c_t(d)=g_{V,t}(d)s_t(d) nondecreasing in depth and per-depth cost λ, min Σλd_t s.t. Σ_t c_t(d_t)≥K has the water-filling condition Δ_t(d_t)=μ for funded tasks and ≤μ for unfunded ones; under single-crossing it is a threshold rule, and tasks with identically zero marginal acceptance are skipped. *Proof:* separable monotone allocation by the Theorem-5 exchange argument with depth in place of compute. ∎

**(iv) Comparison with T7.** Theorem 7′s verifier-assisted ceiling Σ_t g_t is attainable only if g_{V,t}=g_t with unique class (or top-2 class mass 1); otherwise the verifier-assisted ceiling is Σ_t g_{V,t} s_t<Σ_t g_t. The cost is certifying search, not labels: labels are free (train pairs), coverage is not.

**Hybrid correction.** The product form 1−(1−p1)(1−p2) is incoherent for distinct grids; exact-match gives P=p1+p2.

**Frozen notation (T8 hybrid).** Let U=1 denote exactly one verifier behavior class at depth d*, with unique grid v and precision φ=P(v=truth|U=1) (φ=1 under A2). Let w be the TTT top-1 grid and w2 the TTT runner-up; p_T=P(truth=w), p_w2=P(truth=w2) are TTT marginals (working assumption: TTT marginals independent of U, testable). Verifier-first submits (v,w) when U=1 and w≠v; when w=v it falls back to w2; otherwise (w,w2). Coverage is the exclusivity sum E[1{truth=v}+1{truth=w,w≠v}+1{truth=w2,w=v}] (no product term); with multiple classes the top class replaces v at posterior q1. Verifier-first beats TTT-first iff E[1{U}(φ−p_w2)]>0; if verifier coverage is zero (U=0 a.s.) it reduces exactly to p_T+p_w2.

Scope: conditional on A0–A2; a weak DSL makes g_V small and the theorem vacuous; test-time soundness is measured, not assumed. Falsifiers: F1a re-verification (0 violations = PASS); F1b unique-class precision (PASS if φ̂≥0.9, 80% lower ≥0.8, n_unique≥20); F2 hybrid complementarity; F3 acceptance growth by depth.

### 3.11 TTT as amortized Bayesian adaptation

Why are TTT pools plausible? Read ARC-SFT training as approximate posterior inference [13]: the parameters are a prior over task rules, and per-task TTT performs amortized Bayesian adaptation. A LoRA update is a low-rank posterior correction; a PAC-Bayes sketch predicts an interior optimum in rank/steps and larger gains where the prior is weakest. Test-time augmentation is an equivariance prior, valid only where the task rule commutes with the transform; the frozen κ_T proxy records this. It is a sketch, not a bound.

# 4. Accuracy: the offline TTT system and a real public submission

## 4.1 First submission attempt (errored)

We entered [15] with one MIT-0 notebook (Qwen3-4B ARC-SFT, D4×S10 augmentation, per-task LoRA TTT, threshold-DFS decoding, NLL rescoring, coverage selection, schema validation). Submission 57456856 (kernel v8; packet `3774c892…`; S1/aug8) used the 41,400 s budget and 578 s/240-task cap (1156 s/120); hidden size unknown. Kaggle reports COMPLETE at ≈02:35 UTC after ≈2 h 34 min, **no publicScore**, errorDescription *"Your notebook hit an unhandled error while rerunning your code…"*. Cause: rerun branch `ARC_BUDGET_S="41400.0"` parsed by `int("41400.0")` → `ValueError` before try/except. Hardened packet adds int-typed/defensive parses, no-raise validation, a SAFETY cell; probe 57500870 scored 0.00. **We claim no score for 57456856 and report the failure as engineering, not model quality.**

Final hidden-test results (six scored submissions): **S5 16.25** (ref **57681632**, kernel v16; champion), **S6 15.42** (57716141), **S5_ens 15.42** (57752011), **S1 14.17** (57663405), **S5+LegC 14.17** (57785740), CPU probe **0.00** (57500870); plus one invalid-format attempt (57442788) and one rerun error (57456856, Table 9). No score exceeds 16.25. A reference public-notebook replication attempt reached **3/20** on dev20, below our S1 4/20, so the ~30 public-notebook band is external context, not a matched reproduction. The LegC verified-induction hybrid was **net-negative** on the hidden half (−2.08 vs S5) and was rejected. Compute quota was exhausted Oct 31–Nov 1, leaving no GPU for tuning; the Nov 2 final entry uses the best completed version (S5 v16) unless S5b beats 16.25. Half-cost configs truncate at the 578 s cap: these scores are far below the ~30 band, not SOTA.

57456856 exposes no per-task diagnostic; labeled diagnostics come from dev pools (§4.3) and third-party pools (§5).

| date (UTC) | ref | config | LB | status / notes |
|---|---|---|---|---|
| 2026-10-20T14:56 | 57442788 | CPU-fallback insurance | — | invalid format |
| 2026-10-21T00:01:15 | 57456856 | S1 v8, packet `3774c892…` | — | rerun-only `int("41400.0")` ValueError; no publicScore |
| 2026-10-22T00:00:23 | 57500870 | S1 CPU-version probe | **0.00** | COMPLETE, no error |
| 2026-10-26T10:45 | 57663405 | S1 v8 L4, packet `9060264c…` | **14.17** | COMPLETE, no error (≈14 h incl. queue) |
| 2026-10-27T00:00:28 | 57681632 | **S5 v16 L4**, packet `6211b694…` | **16.25** | COMPLETE, no error — **champion** |
| 2026-10-28T00:00:27 | 57716141 | S6 dual-config v22 | **15.42** | COMPLETE |
| 2026-10-29T00:02:23 | 57752011 | S5_ens two-model v23 | **15.42** | COMPLETE |
| 2026-10-30T00:02:21 | 57785740 | S5+LegC hybrid v25 | **14.17** | COMPLETE — LegC net-negative |
| 2026-10-31 | — | (submission missed) | — | session gap; no ref |
| 2026-11-01T00:10:23 | 57854697 | S5b v24, packet pending | pending | PENDING at freeze — last S5-family probe |

Table 9. Submission ledger (all public-LB attempts; errors and misses included).

Hidden-set size remains unknown; runtimes are queue time, not a size signal. Six scored entries; all scores are public-LB hidden-half results. Table 8′s dev20 scores are public-eval-subset measurements, not hidden scores.

## 4.2 Hidden-test budget calibration

Hidden size is unknown (120–240); for the 240/259 sample shape at 11.5 h, the four-worker budget is 41,400 s and the fair per-task cap is 578 s at 240 tasks (1156 s if 120; 9.6/19.3 min), including a reserve. The cap truncates attempts. Dev2 timings: ≈220 s/task (202.51/238.12 s). Scaling to 240 tasks on four workers gives ≈3.7 h of TTT — 2.6× per-task headroom, ~7.8 h unused wall. Slack absorbs harder tasks, model load (106.9 s) and worker skew. Processed-task count and truncation are not observable

from the rerun interface. Figure 9 reserves this calibration and the §4.3 dev compute-accuracy points. Theorem 5 predicts compute beyond a coverage plateau is wasted; the dev20 ladder tests this (Table 11; EXPLORATORY/PENDING).

### 4.3 Compute-normalized dev evidence

On this single 20-task subset (no CI claim), the all-config union (incl. S5_ens) reaches 6/20 (30%) versus 4/20 for S1; four-config S1/S2/S5/S6 union 5/20. Dev2 pilot: 466.7 s wall, 106.9 s load, 15 candidates for 2/2 inputs, oracle 1/2 — plumbing evidence, not accuracy.

| config | dev20 (task) | dev20 (input) | final LB | notes |
|---|---|---|---|---|
| S1 (hardened) | 4/20 = 20.0% | 6/32 = 18.8% | **14.17** | decode mean ≈235 s/task; 4 tasks capped |
| S2 (aug16) | 2/20 = 10.0% | 5/32 = 15.6% | — | decode up to 618 s; truncation |
| S5 (single) | 3/20 = 15.0% | 5/32 = 15.6% | **16.25** | champion |
| S6 (v13; selection-bug) | 2/20 = 10.0% | 5/32 = 15.6% | **15.42** | pass@2 flagged; 4/20 oracle, usable |
| all-config union | 6/20 = 30.0% | — | — | cand.-union incl. S5_ens; four-config 5/20 |

Table 8. dev20 sweep (eval20: 20 tasks / 32 inputs; L4×4; hardened packets).

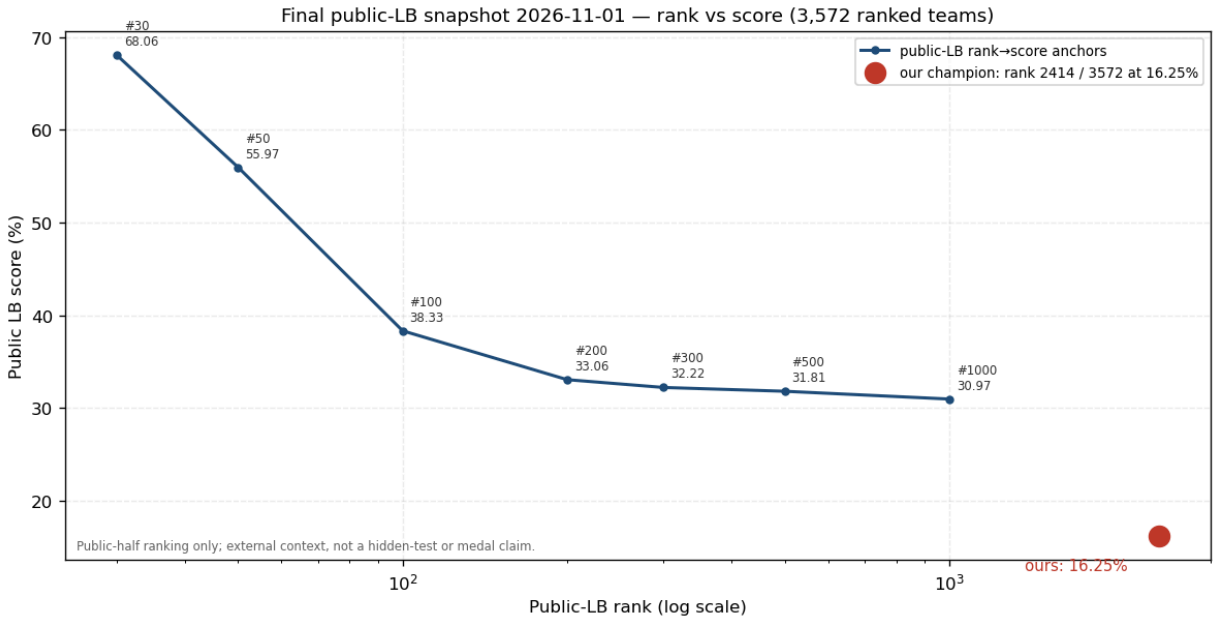



Figure 9. Final public-LB snapshot (2026-11-01T00:13Z): rank↔score anchors and our champion at rank 2414 / 3,572 (score 16.25). Public-half ranking only; external context, not a hidden-test or medal claim.

**Table 11. Public-LB snapshot 2026-11-01T00:13:02Z.** 3,572 ranked teams; leader 88.06. Our row: `rank 2414 · "wen Chen" · TeamId 17156552 · score 16.25 · submissions 8 · last 2026-10-30 00:02:21` (three teams tied at 16.25, ranks 2412–2414; 67.5th percentile from top). Anchors: r30 68.06, r50 55.97, r100 38.33, r200 33.06, r500 31.81, r1000 30.97. The public LB displays our **best** score (16.25), not the last submission (v25 = 14.17): standings use best-of-submissions semantics, so the Nov 2 entry is only a safety copy for the last-submission rule. The ~30–31 notebook band is ≈ranks 500–1000 (top 14–28%); 16.25 is far below it — public-half ranking only; no medal/SOTA claim.

### 4.5 Verifier experiments: bounded-DSL results

We tested an AWM-style verifier-first search (LegC DSL; T0 identity, T1 atoms, T2 composition) on public eval120. The depth-2 engine accepted **0/120** programs: unique coverage 0%, attempt-1 precision undefined, **certificate fraction 120/120 impossibility**, 35.05 CPU-min total (0.292 CPU-min/task; serial; wall 2103 s). F1a is PASS but vacuous (nothing accepted to re-verify); F1b is INCONCLUSIVE (no unique classes); F2′s pre-registered verdict is **FAIL** (hybrid = TTT top-2; on the exploratory 20-

task subset S1 4/20 = 20.0% vs verifier-only 0%, H–T=0, 80% CI [0,0]); because verifier coverage is zero, the complementarity hypothesis is untestable in interpretation and the hybrid reduces exactly to TTT top-1+runner-up, consistent with T8′s coverage premise failing on AGI-2. F3 **FAILS at measured depth 2** (acceptance flat for all 120). **Depth-3 subset (20 dev20/joint ids, beam 200): 0/20 certified, 20/20 impossibility certificates (1765.6 s serial; T3 executed on all tasks); not extrapolated to eval120; d3 remains not measured at eval120 scale.** Learned-rule families (0/120), DeepSeek-chat (0/20), and DeepSeek-reasoner (0/10; all completions token-truncated) give nulls; the reasoner null is **infrastructure-limited, not capability evidence**. The n=12 training control yields **1/12 unique solve (0520fde7)**, confirming the engine can certify reachable programs; it is small and not used for a rate claim.

| run | n | accepted | unique | empty-certificate | note |
|---|---|---|---|---|---|
| eval120 | 120 | 0 | 0 | 120/120 | 35.05 CPU-min serial; wall 2103 s |
| train control | 12 | 1 | 1 | 11/12 | unique 0520fde7; 1/12, no rate claim |
| learned rules | 120 | 0 | 0 | — | 0.3 min |
| LLM chat | 20 | 0 | 0 | — | DeepSeek-chat k=4 |
| LLM reasoner | 10 | 0 | 0 | — | token-cap truncation: infrastructure null |

Table 14. Verifier search runs (public eval120; labels used for scoring only; not a competition entry).

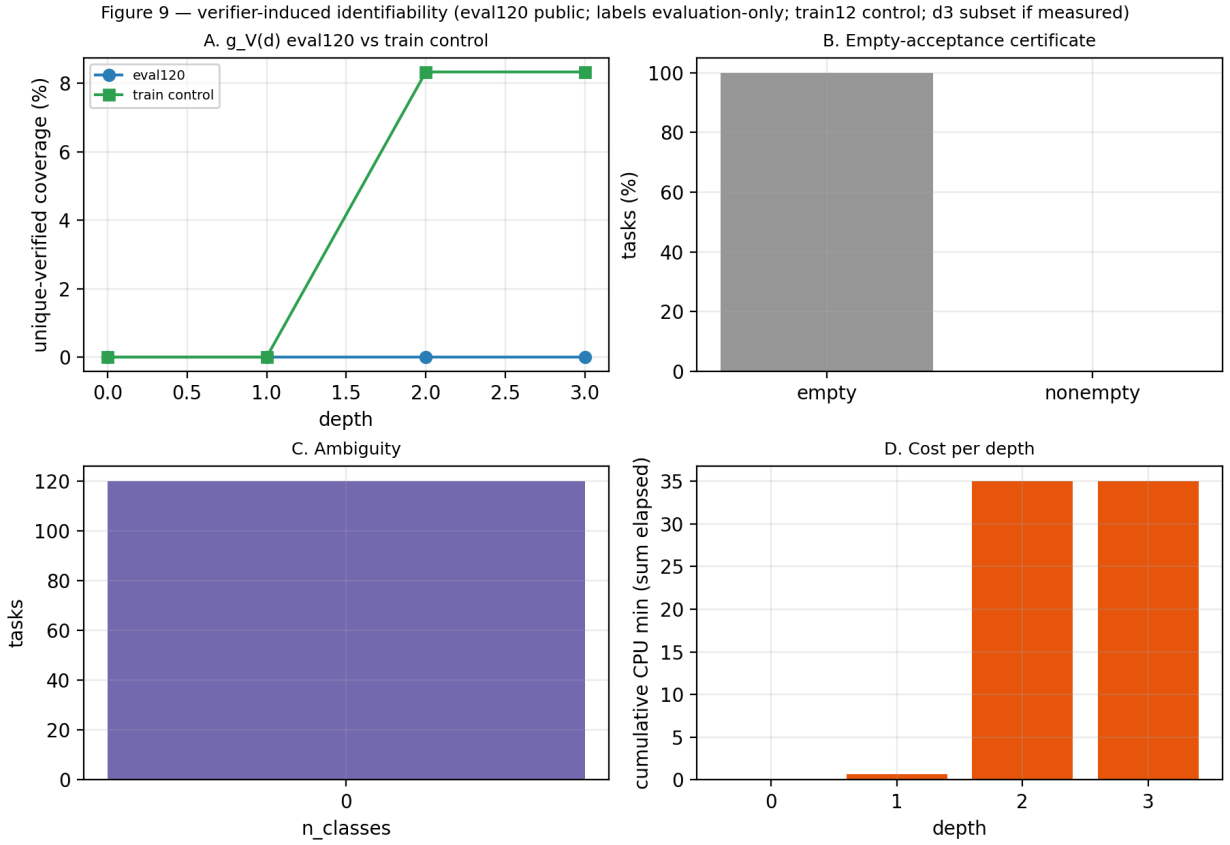



Figure 9. Verifier search on public eval: (A) `g_V(d)` for eval120 vs the n=12 training control; (B) impossibility-certificate fraction by depth; (C) class ambiguity; (D) cost per depth. Public eval, labels scoring-only, not a competition entry; d3 measured only on the 20-task subset, not extrapolated to eval120.

# 5. Empirical study: selection rules on third-party candidate pools

## 5.1 Sources, reproduction gates, and protocol

Released third-party logs bound the study: gains are pool-conditioned; sources are frozen and a failing gate excludes one. Table 3 gates: B's coverage summary (29 inputs; oracle 9/29; top-2 8/29) and A's artifact (137/400 solved; 80/400 pass@2) reproduce exactly. Rules: duplicate, argmax, top2, cal_indep, coverage_optimal, oracle_any2 (ceiling), deployed. Splits are task-level (B 9/11 tasks, 13/16 inputs; A 400/400); paired bootstrap over tasks (2,000 resamples, seed 0) [14] gives 80%/95% CIs; no refits.

| source | gate | published | our reproduction | pass |
|---|---|---|---|---|
| B | coverage summary | 29 inputs; oracle 9/29; top-2 8/29 | exact, 0 per-input mismatches | yes |
| A | upstream script, 2000 steps | solved 137/400; ≤2 guesses 80/400 | exact via bundled artifact | yes |

Table 3. Reproduction gates (source-published metric vs our re-derivation).

## 5.2 Pilot evidence on Dataset B

B is the only public candidate-dump notebook; 20 tasks, pilot-only, holdout 16 inputs/11 tasks. Oracle 31.25% (5/16); every deployable rule (duplicate, argmax, top-2, cal_indep, coverage) also 31.25%, indistinguishable (80% CI [18.75,43.75]); headroom 0. Deployed fallback 25.00%.

## 5.3 The large replication and the simulated stress test

A is claim-bearing: top-2 covers 80/400 (20.00%) vs 137/400 any-guess (34.25%). The result is a null: correctness is predictable but score-dominated. Missed truths are far from the top (rank median 12). `DOWNGRADE_TO_ANALYSIS`; deployed fallback 19.25%; CompressARC [6] is ARC-AGI-1-era (hash-level pools).

C isolates the estimator; a sanity check, not evidence: calibrated rules reach the oracle 41.67% (+18.33 pp, 95% CI [11.67, 25.0]; gate slope 0.025).

| rule | B: ARC-AGI-2 TTT pools (16 inputs / 11 tasks) | A: CompressARC (400 tasks) | C: simulated (120 tasks) |
|---|---|---|---|
| oracle_any2 (ceiling, not a rule) | 31.25 [18.75, 43.75] | 34.25 [31.25, 37.50] | 41.67 [35.83, 47.50] |
| duplicate | 31.25 [18.75, 43.75] | 18.50 [16.00, 21.00] | 16.67 [12.50, 20.83] |
| argmax1_argmax2 | 31.25 [18.75, 43.75] | 19.50 [17.00, 22.00] | 23.33 [18.33, 28.33] |
| top2_consensus | 31.25 [18.75, 43.75] | 20.00 [17.50, 22.75] | 21.67 [16.67, 26.67] |
| cal_indep | 31.25 [18.75, 43.75] | 18.00 [15.50, 20.75] | 41.67 [35.83, 47.50] |
| coverage_optimal | 31.25 [18.75, 43.75] | 17.75 [15.25, 20.25] | 41.67 [35.83, 47.50] |
| deployed (gate-selected) | 25.00 [12.50, 37.50] | 19.25 [16.75, 21.75] | 41.67 [35.83, 47.50] |

Table 2. Frozen two-attempt rules on the three datasets (pass@2, %; 80% bootstrap CI in brackets; seed 0, 2,000 resamples; two decimals).

Cells are pass@2 from the frozen artifact `analysis/results/selection_benchmark.json` (sha256 `6c16abfef7f0efe3a1a6bbe9f6251c9aeef1893ee7b0f44fdfdb9c2eb34ab71a`; 80% CIs). C is simulated, not evidence about real solvers.

**Score sufficiency** explains the null: score ordering saturates recoverable coverage, so selection is not the bottleneck — generation is; re-ranking never exceeds top-2's 20.00%, the 14.25-point headroom is real but not separable from exposed metadata. Falsifiers: (F1) a calibrated rule beating the solver by ≥+1.0 point (80% CI lower >0); (F2) metadata exposing oracle-gap candidates; (F3) a multi-decoder union exceeding noise.

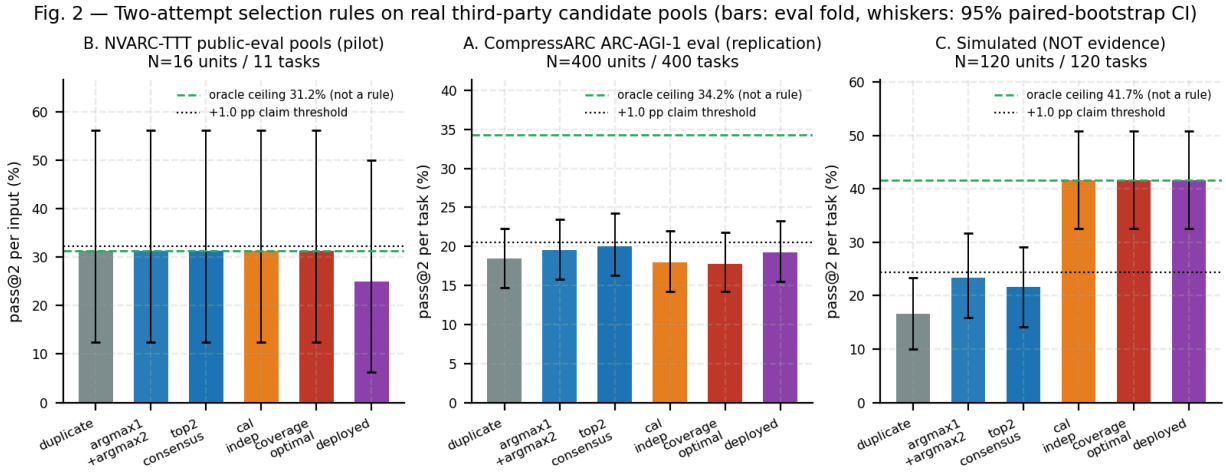



Figure 2. Rule comparison with bootstrapped CIs: pass@2 and paired Δ vs the published baseline, per dataset (B pilot; A replication; C simulated). Oracle coverage is a ceiling, not a rule; bars are the pre-registered 80%/95% intervals; the +1.0-point line applies to the claim-bearing dataset only.

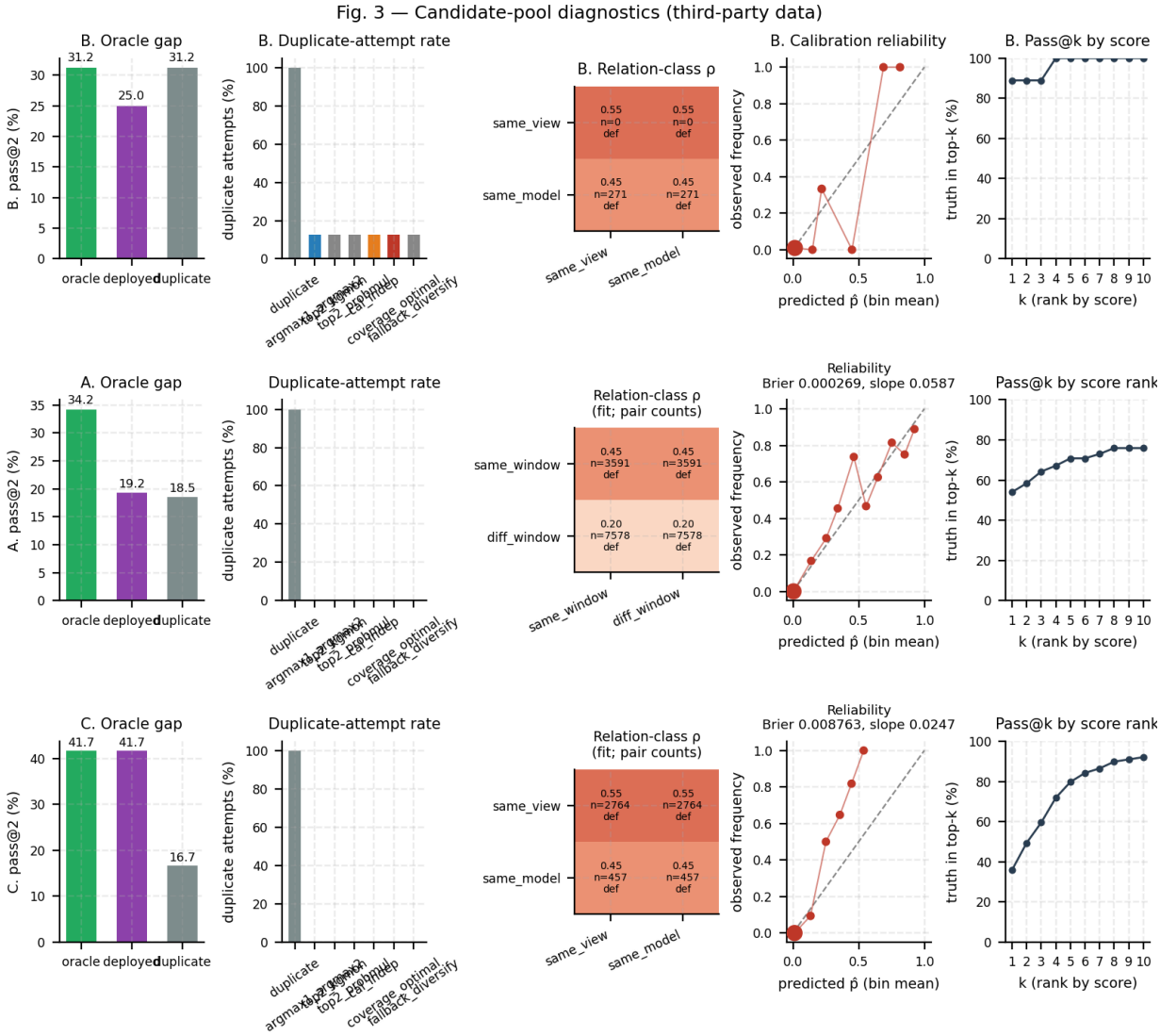



Figure 3. Pool diagnostics: oracle gap, duplicate rates, relation-class ρ heatmap with pair counts, and calibration reliability for B/A and C.

### 5.4 Large-scale numerical verification of T6/T7

Two cloud-box runs check the theorem implementations at scale. **T6:** 100,000,000 synthetic instances (`bound_sim_gpu_100m.json`) produce 30,063 floating-point excesses, **all below the pre-registered 1e-6 tolerance (zero violations)**; maximum excess `1.12e-7`, mean bound-cap gain .209, 28.3 s. **T7:** 2,000,000 tasks (`alloc_sim_gpu_2m_v2.json`) give mean `δ=.260316`, label-free limit .739684, verifier ceiling 1.0, and gap `.260316 = mean δ` — budget-invariant across the C=50–5000 ladder; water-filling beats uniform by 3.5–8.8× at every rung, 20.1 s. These validate the implementations, not ARC performance. Canonical summary: `analysis/results/gpu_validation.json` sha `4970955c545f460d…` (+ `gpu_validation.md` `1c6253c4f022b91d…`). Artifacts: `analysis/results/gpu_sim/bound_sim_gpu_100m.json` sha `b1117f3812ab5200…` and `alloc_sim_gpu_2m_v2.json` sha `e39cd282796366ae…` (superseded alloc v1 `629de32d…`), both from `code/lab/gpu_theory_sim.py` (published in the MIT-0 repo; same file for CPU/GPU). Commands: `python code/lab/gpu_theory_sim.py bound --n 100000000 --out out/bound_sim_gpu_100m.json` (seed 7; K=8 rank categories; acc~U[0.1,0.9]; τ prior p_j∝j^{−α}, α~U[0.5,3.0]) and `python code/lab/gpu_theory_sim.py alloc --n-tasks 2000000 --budget 50 --out out/alloc_sim_gpu_2m_v2.json` (seed 11; τ=exp(N(0,0.8)−0.5) clamped [0.05,20]; identifiable fraction 0.35; δ∈[0.2,0.6] for unidentifiable, 0 otherwise; g(c)=1−exp(−c/τ); bisection 80 iters). Device: NVIDIA TITAN Xp 12,288 MiB, driver 580.105, torch 2.5.1+cu118, CUDA 11.8; 56 vCPU / 94 GB RAM; 9.16 TFLOPS FP32 (4096³ matmul ×20); runtimes 28.3 s (bound) and 20.1 s (alloc).

| check | n | result | time |
|---|---|---|---|
| T6 bound | 1e8 | 0 violations at 1e-6; max excess 1.12e-7 | 28.3 s |
| T7 allocation | 2e6 tasks | gap = mean δ = .260316; water-filling 3.5–8.8× uniform | 20.1 s |

Table 15. GPU-scale verification.

## 6. Data characterization and leakage audit

We characterize ARC-AGI-2 data [20]: 1,000 training tasks (3,232 pairs, 1,076 inputs) and 120 evaluation tasks (358 pairs, 172 inputs); the repo mirror (167 inputs) is not used for scores. A SHA-256 audit finds zero shared tasks, pairs and input grids between training and evaluation.

Heuristic baselines solve 1.6% training, 1.5% holdout and 0% eval; eval grids are larger (21.95 vs 14.67, p<0.001) and 54.5%/57.5% of train/eval tasks are same-size rewrites.

| quantity | value | artifact |
|---|---|---|
| train / eval tasks | 1,000 / 120 | §1 |
| shared tasks / pairs / input grids | 0 / 0 / 0 | §12 |
| trivial-heuristic solve: train / holdout / eval | 1.6% / 1.5% / 0% | §13 |
| max grid dimension: train vs eval | 14.67 vs 21.95 (p<0.001) | §11 |
| solvability tiers A/B/C | 28 / 56 / 36 (uncalibrated) | §15 |

Table 6. Data-characterization summary (public ARC-AGI-2; unless noted).

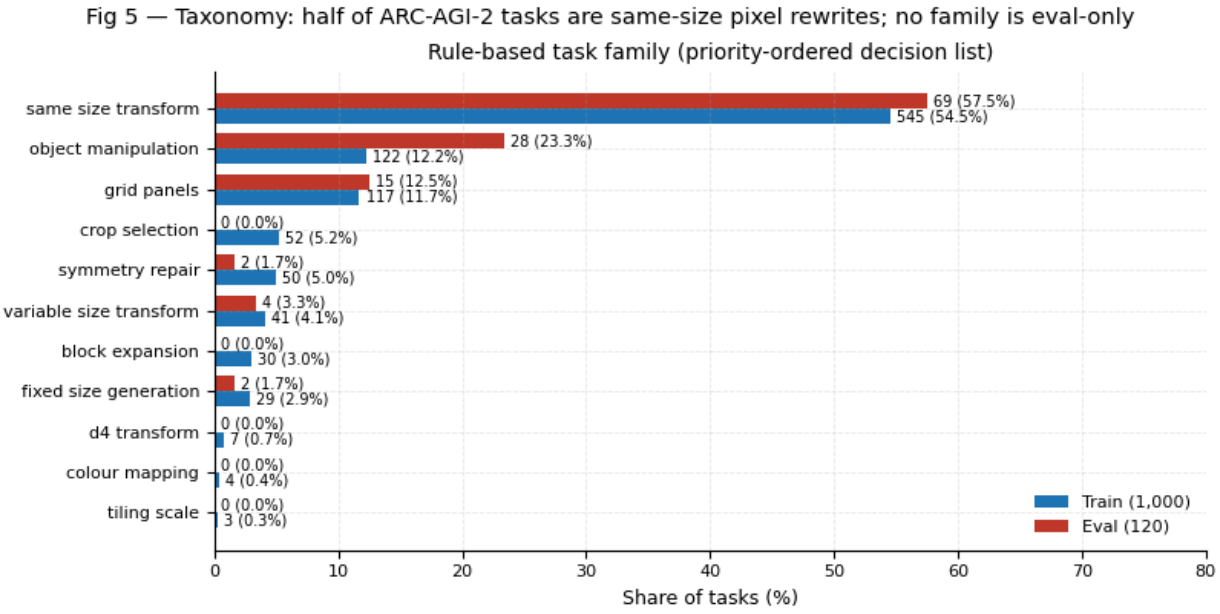



Figure 5a. Rule-based task-family distribution for 1,000 training and 120 public-evaluation tasks (analysis/report.md §9). Same-size rewriting dominates both; no family is evaluation-only.

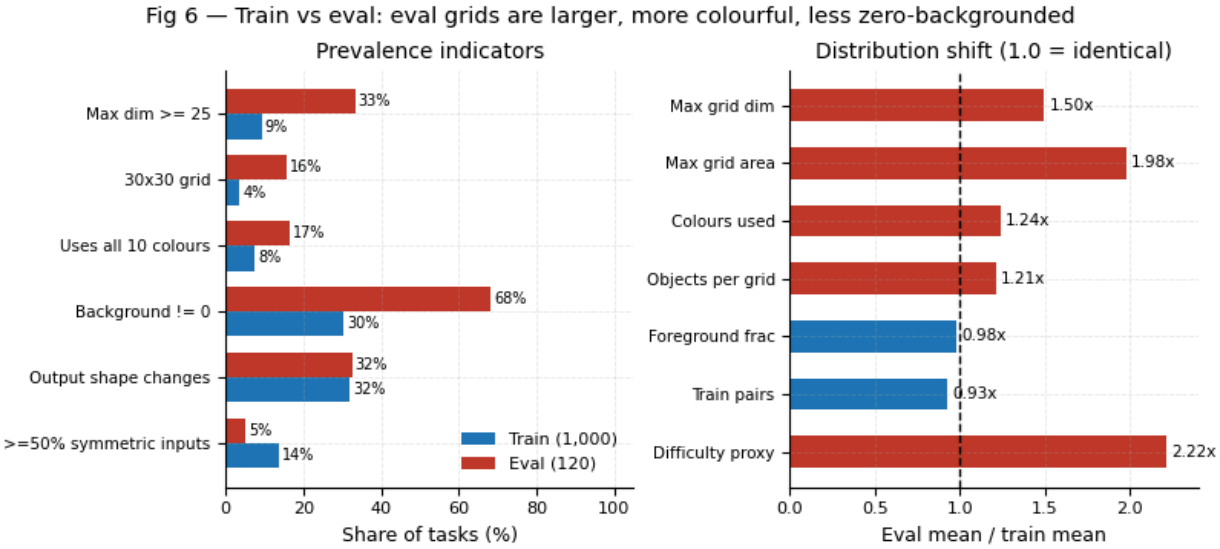



Figure 5b. Train-vs-evaluation shift: prevalence indicators (left) and eval/train mean ratios (right) for size, palette, demonstrations, and background (analysis/report.md §11).

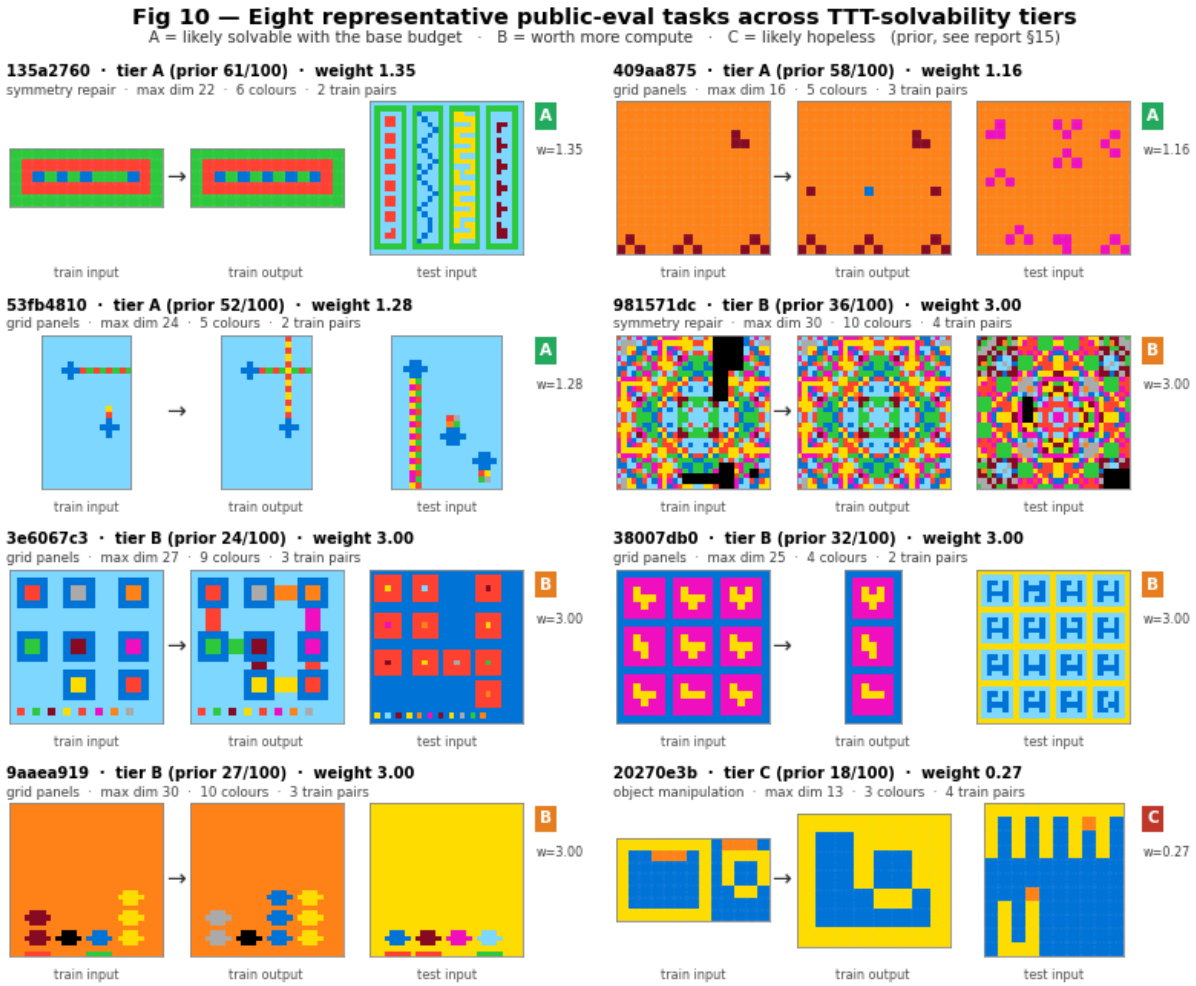



Figure 6a. Eight representative public-evaluation tasks across prior tiers with budget weights (analysis/report.md §15); tiers are an uncalibrated prior, not a solve rate.

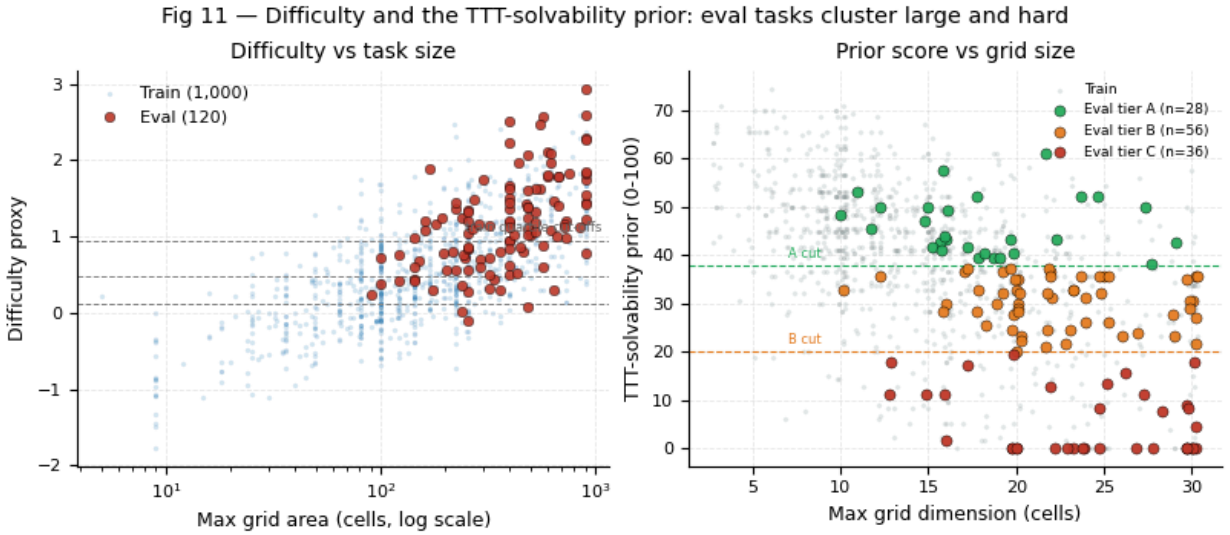



Figure 6b. Difficulty proxy and the uncalibrated TTT-solvability prior versus task size (analysis/report.md §15–16).

## 7. Completeness: engineering and reproducibility

The repository implements loader, augmentation, threshold-DFS decoding, rescoring, aggregation, coverage rules (§3), TTT and orchestration; it runs offline from a kernel mount. Released under

MIT-0; third-party components retain licences (ARC-AGI-2, Qwen3, CompressARC, arc-dsl, Unsloth/peft). Notebook methods are clean-room re-implemented and cited; a NOTICE records attributions. The `LICENSE` file must switch to SPDX `MIT-0` before release.

GPU fixes: Triton needs an explicit `TRITON_PTXAS_PATH=/usr/local/cuda/bin/ptxas` (missing ptxas → zero candidates), and Hugging Face loading needs offline weights (`local_files_only`/mount). The packet pins source/hash and forces deterministic fallback; Validation: 190 passing CPU tests (~26 s); a schema-valid fallback: the 167-input packet ran in 8.3 s on Kaggle, while the official 172-input repackaging measured 5.4 s locally and scores 0/120 — a safety net, not accuracy; TTT-path validation with a tiny Qwen model (dialect guard, LoRA, 2 steps, 8-view decode, selection, schema); and a CompressARC harness that re-derives bundled predictions and exports hash-level pools for Dataset A.

| component | validation evidence | status |
|---|---|---|
| CPU test suite | 190 tests pass (~26 s) | measured |
| CPU fallback submission | repo 167-input packet 8.3 s on Kaggle; official 172-input repackaging 5.4 s local, 0/120 (safety net) | measured |
| TTT path (tiny model) | tokenizer guard, LoRA, decode, selection, schema OK | measured |
| GPU dev2 pilot | 2 tasks, 15 candidates, per-task walls 202.5/238.1 s, load 106.9 s | measured |
| Engineering fixes | ptxas/HF-offline guards; void stale-packet run root-caused | measured |
| First submission attempt | 57456856 errored (rerun-only budget parse); CPU-probe retry (57500870) publicScore 0.00 | measured failure |
| First scored hidden run | **57663405**, S1 L4, COMPLETE **publicScore 14.17** (≈14 h incl. queue) | measured (scored) |
| Best scored hidden run | **57681632**, S5 L4, COMPLETE **publicScore 16.25** (current best; +2.08 pp over S1) | measured (scored) |
| S6 dual-config rerun | Oct 28 submission | PENDING |
| CompressARC reproduction | 137/400 solved; 80/400 pass@2 | measured |

Table 5. System and reproducibility matrix.

## 8. Public-solver landscape and what transfers

Table 4 has ten notebooks [16–19]; nine cluster at 29.7–31.4. TRM [5], SOAR [7], deep ensembles [11] are not evaluated.

Saved artifacts are partial reruns (vanductan 5/120; 8/172; others 2–4/120).

| notebook (family) | displayed | claimed | mechanism | transferable element |
|---|---|---|---|---|
| mayankraaz | 31.39 | 35.97 | triple-pass pooled TTT, controller, LOO-DSL | cheap-first ordering; pass-A invariant |
| yuvrajlamba01 | 31.39 | 54.31% | TTT + deterministic symbolic solver | monotone symbolic filler |
| rabari9999 | 30.97 | n/a | v38 perf patch clone | GPU-side log-softmax patch |
| vanductan | 30.97 | n/a | perf-patched research dump | candidate-pool logging (Dataset B) |
| mikelou1 / anthonyxlyu | 30.14 | 33.89 | v38 minimal perf patch | algebraically identical speed patch |
| woominyo | 29.72 | n/a | perf patch + interleaved ensemble | attempt ordering via interleaving |
| dmitriigluzdov | 29.17 | n/a | +6 transformer layers | negative result — capacity expansion |
| delongyang | 15.69 | n/a | 27B FP8 zero-shot | model-size/throughput trade-off |
| cprs | 30.56 | 80%+ target | TTT + LLM program-synthesis pre-pass | verified-program candidate source |

Table 4. Public-solver landscape (dated reading 2026-10-18; displayed scores are secondary-source context).

## 9. Limitations, ethics, and reproducibility

The largest limitation is that our solver has **six scored hidden submissions** spanning **14.17–16.25** (champion S5 16.25), versus **20.0% task-level** on the 20-task public-eval subset — S1/S5 gaps ≈5.8/≈3.75 points consistent with half-cost truncation and subset noise, not distributional. Compute quota was exhausted Oct 31–Nov 1, leaving no GPU for further tuning; the ~30 public-notebook band is external context only. The first rerun (57456856) errored after ≈2 h 34 min with no publicScore; probe 57500870 scored 0.00. No per-task hidden observability exists, so variance cannot be reported. The selection claim remains a third-party-pool study with narrow conclusions; B is tiny (29 inputs/20 tasks), A overlaps ARC-AGI-2 training, C validates only estimator/implementation.

**D1/D2 caveats.** The D1 inputs are empirical consistency proxies, not true posteriors; metadata are partial (third-party dumps; pilot/CompressARC lack shape/view metadata), so `I` is a lower bound and the certificate may overstate achievable coverage; hidden-task transfer is an explicit exchangeability assumption, not a guarantee. Label gaps: `/tmp/dv15/logs` and `logs/pools` were missing, so archived dev20 paths were used; dev20 has only 16/32 units and wide CIs; S6 is selection-bug-affected; S5_ens is a 53-task superset; the +8.11 pp S5_ens headroom is selectable, not realized (hidden 15.42 < S5 16.25). T6 is a **negative result** plus a conditional certificate: certificates are vacuous on all pools ($I \geq 2T^2$); gain=T holds on dev20 S2/S5/S5_ens (ratio 1.00) while CompressARC loses −2.25 pp; Prop. 6f is tight only under score sufficiency. D2 is definitional, not T7 evidence; T7′s only support is the hidden-score ladder, **consistent not conclusive** (n=6; truncation; S6 bug; 53-task superset; packet/config differences; unknown hidden size). **Verifier caveats.** Verifier results are public-eval diagnostics: the bounded-DSL engine accepted 0/120 (impossibility certificates for the bounded class, not all programs); F1a is vacuous, F1b/F3 are inconclusive/fail-at-depth, F2 is untestable with zero verifier coverage; the n=12 train control is small; LLM nulls are token-cap infrastructure-limited; labels are used for scoring only and never inside the search; no hidden-test or SOTA claim. Theory is bounded: coverage identities are exact, but $\hat{p}$ and $\hat{\rho}$ are estimates; the gate is a guard, not guarantee; TTT is a sketch.

All data are public (ARC Prize data, notebook outputs, CompressARC artifacts); no personal data, credentials or hidden-test information is used. Third-party artifacts are used with attribution; no third-party code was copied. Reproducibility depends on Appendix B artifacts and on the repository reaching a single consistent licence before release.

## 10. Conclusion

Coverage is the right formalization for two-attempt selection: independent correctness makes top-2 optimal, correlated errors make a calibrated pair rule provably better, and Theorem 4 gives the correlation threshold. The measured answer is a bounded negative: on real public pools the solver's score ordering saturates recoverable coverage. On 400 CompressARC tasks top-2 covers 20.00% while the coverage rule covers 17.75%, against 34.25% oracle — 14.25 points exposed metadata cannot reach. The verdict is `DOWNGRADE_TO_ANALYSIS`; selection is not the bottleneck, candidate generation/coverage is. The information view sharpens the result: Theorem 6 bounds any metadata rule by `p1+p2+min{T,sqrt(I/2)}`, so the 34.25% vs 20.00% gap is an identification deficit, not a ranking failure; the certificates are vacuous on every pool ($I \geq 2T^2$), so T6 is proven rather than empirically separated; on dev20 S2/S5/S5_ens selection recovers the tail T exactly (ratio 1.00), while CompressARC loses −2.25 pp, and the S5_ens headroom is selectable but unrealized. Theorem 7 (conditional) makes the separation explicit: verifier-free compute leaves `ΣΔ_t` unchanged while identifiable coverage scales; the hidden-score ladder (16.25 > 15.42 = 15.42 > 14.17) is consistent but confounded, not conclusive. Theorem 8 adds the verifier route: unique accepted classes restore identifiability, impossibility certificates bound search depth, and the bounded-DSL engine certifies 0/120 eval tasks (train control 1/12), so verification cannot close the gap without reachability. Transferables: the formalization, information bound, protocol, benchmark, data audit, implementation, real attempt and record — plus falsification targets: held-out gain outside the certificate, large $\hat{I}$ with zero gain, a vacuous bound with zero gain, or a monotonicity violation.

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*All web sources accessed 2026-10-18 (refs 26–30: 2026-10-20); arXiv identifiers resolve at `arxiv.org/abs/<id>`.*

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## Appendix A — Pre-registration (condensed from `research/design/selection_protocol_v2.md`)

- **Sources frozen before evaluation:** B = `vanductan/arc-agi-2-candidate-dump` (pilot; 29 inputs/20 tasks — no other public candidate-dump notebook was found); A = CompressARC bundled predictions (claim-bearing; 400 ARC-AGI-1 eval tasks, fit on 400 train; hash-level candidates).
- **Reproduction gates:** B must reproduce 29 units, oracle 9, top-2 8; A must reproduce 137 solved and 80 pass@2 at 2,000 steps. A failed gate excludes the source; it does not trigger rule changes.
- **Units and splits:** B unit = test input, tasks split 9/11 dev/holdout by task hash (13/16 inputs); A unit = task, source-defined split (400/400).
- **Frozen scalar scores:** B = Σ of the eight per-augmentation scores (lower better); A = logaddexp-accumulated pick score (higher better).
- **Rules:** duplicate, argmax1_argmax2, top2_consensus, cal_indep, coverage_optimal, oracle_any2, deployed (selected on fit).
- **Features for $\hat{p}$ (fixed order):** score rank percentile, score z-score, log(1+agreement), cluster mass, is-top-cluster, shape consistency (B), cross-model count; A adds log first-step and pick-type diversity.
- **$\rho$ classes:** same_candidate ($\rho$=1, never paired), same_view, same_source, cross_source (pooled only; no cross-source claim).
- **Statistics:** paired bootstrap over tasks, 2,000 resamples, seed 0; 80% and 95% CIs; claim iff $\Delta \geq +1.0$ point and 80% CI lower > 0 on the claim-bearing holdout.
- **Verdict:** Dataset A does not meet the pre-registered margin — coverage_optimal 17.75% vs argmax1_argmax2 19.50%, margin −1.75 pp (required: ≥+1.0 point with 80% CI lower bound >0) — so the selection claim is `DOWNGRADE_TO_ANALYSIS` and the deployed rule is the no-calibration fallback. Dataset C is simulated and therefore never enters the claim.
- **Deviations:** any change is a new protocol version committed before the affected holdout is read; both versions reported.

## Appendix B — Evidence registry and per-number provenance

All numbers map to rows in `paper/evidence_table.md` (Plan C). Primary artifacts: `analysis/results/selection_benchmark.json` (Table 2 and Fig. 2; command `python analysis/selection_benchmark.py`); `logs/arch-theorist/external_reproduction_gate_arc2_dump.json` (Dataset B reproduction gate); `code/out/compressarc/verify_bundled_eval.json` (Dataset

A reproduction gate); `analysis/report.md` (data characterization); `analysis/results/saved_artifact_scores.json` (landscape); `logs/kaggle_freeze.md` and `logs/kaggle_runs.md` (platform gate); `code/docs/RESULTS.md` (CPU baselines and system validation). The theorem and the pre-registration are documented in `research/design/theory.md` and `research/design/selection_protocol_v2.md`, released with the paper bundle. GPU end-to-end execution and six scored leaderboard submissions are measured (Tables 8–9); no hidden-test data were accessed locally and no per-task hidden outcomes are available.

## Appendix C — Platform-gated access log (summary)

Thirteen Kaggle runs recorded between 14:00 and 15:06 UTC on 2026-10-18, all CPU-only: five environment/mount/tokenizer probes, three fallback-packet dry runs, five TTT-packet smoke runs. GPU entitlement was requested (`NvidiaL4`, `NvidiaTeslaT4`) and not granted; competition attach was rejected pending rules acceptance, which requires phone verification. Zero Kaggle GPU-hours were consumed; the frozen handoff bundle contains packet checksums and a resume procedure that regenerates packets before pushing. No hidden-test data was accessed and no GPU number appears in the paper. **V4 update (2026-10-21):** the competition gate cleared; kernel v8 dev2 pilot measured on 4×L4 (§4.3). Submission **57456856** reran (2026-10-21T00:01:15Z) and errored before compute: the rerun branch set `ARC_BUDGET_S="41400.0"` and the run cell's `int("41400.0")` raised `ValueError`; Kaggle's ≈2 h 34 min was queue time. Hardening (int-typed budgets, no-raise validation, sanitization, SAFETY cell, rerun-simulation gate) landed; probe 57500870 scored 0.00; L4 refs 57663405=14.17, 57681632=16.25; S6 15.42.

## Appendix D — Reproduction commands

```
# data analysis, split, baselines, allocation, figures (CPU, 169 s)
bash analysis/run_all.sh

# selection benchmark on third-party pools (CPU; needs the downloaded dumps)
python analysis/selection_benchmark.py

# repository tests and CPU baselines
python -m pytest code/tests -q
python code/compressarc_harness.py verify-bundled --out code/out/compressarc/
verify_bundled_eval.json
```

## Appendix E — Diversification-threshold derivation, P1/P2 protocol, and falsifier

Theorem 4 and its corollaries are stated in §3.4; the full derivation, edge cases, and the exact inversion of the threshold are in `research/design/theory.md` §4.2. The empirical protocol is frozen in `research/design/selection_protocol_v2.md` §6d and implemented in `research/design/diversification_threshold.py` (self-test covers theorem arithmetic, admission/bootstrap mechanics and pilot flagging). On `pool-v1` logs from our own GPU runs: unit = (task, test input); task-level fit/holdout split by `sha256(task_id)` parity; $\hat{p}$ from the frozen dev-fit logistic calibration and $\hat{\rho}_c$ from the method-of-moments co-failure estimator per relation class (same_view, same_model_cross_view, score_tied, cross_model), clipped to $[-0.95, 0.99]$; $\delta=0.05$; pairs cap at the top 32 by $\hat{p}$. Rules compared on holdout units: top-1 duplicated, threshold-diversify (admit iff $\hat{\rho}_c < \rho^* - \delta$), and the ungated coverage-optimal contrast. Pre-registered predictions: **P1** admitted pairs beat their duplicate counterpart (paired $\Delta>0$, 80% CI>0); **P2** rejected (score-tied) pairs do not ($\Delta\leq 0$); **P3** predicted-vs-realized gain slope in $[0.7, 1.3]$. Falsifier: failure of P1, P2 or P3 falsifies the estimator/relation-class instantiation, never the algebra. The dev20 pools (65 base tasks / 133 task-pools) were analyzed; exact-match exclusivity makes P1/P2/P3 untestable (theory §4.8): every class has $\hat{\rho}<0$, so Table 10 is a prose check and Fig. 9 is omitted.

## Appendix F — Official competition data note

The official evaluation files (`data/comp/`, sha256 prefixes `e7c62a4b…` / `84be4f4f…`) contain 120 tasks and **172** test inputs; the repo mirror distributed with the public data has 167. Five tasks differ: four add an extra test input (inserted first, so attempt order shifts), `faa9f03d` also differs in its train pairs, and `d8e07eb2` has a different train-pair output. Repo-mirror scores are therefore not official scores, and all official numbers in this paper are computed by `analysis/official_scoring.py --recompute-all`. The local `arc-agi_test_challenges.json` file (240 tasks / 259 inputs) is an exact content duplicate of 240 public training tasks and is a local dev sandbox only; it is never compared to the hidden leaderboard. Rerun budget: 4 workers × 12 h for ~240 tasks / 259 inputs gives 12.0 min per task / 11.1 min per input (10.8 / 10.0 min with a 10% margin). Dataset B's selection benchmark attached truth from the repo mirror; the one differing task present in B (`abc82100`) has no candidate matching any repository or official truth for its decoded input, so no rule covers it and the B row is unchanged (checked in `paper/official_b_truth_check.md`).

## Appendix G — Theorem 5 derivation, dev20 protocol, and falsifiers

Appendix G / Table 11 (reserved): Theorem-5 derivation (Jensen, KKT/exchange, single-crossing, saturation); dev20 ladder S1/S2/S5/S5_ens ⟶ per-task wall, candidates, coverage, marginal gains by $\hat{p}$ proxies; PENDING/EXPLORATORY.

## Appendix H — Exploratory cross-source note

An exploratory check on the partial saved artifacts (task-level 50/70 dev/holdout split, source quality estimated on dev only) found no conversion: on the 70-task holdout the best dev source solves 1/70, best-source attempt-1 plus cross-source attempt-2 also solves 1/70, while the union oracle is 8/70. Cross-source diversification does not convert the observed pool headroom on these partial artifacts. This is an appendix note only; it is not part of the pre-registered study, the artifacts are partial eval-mode runs rather than leaderboard reruns, and source quality was estimated on the dev split only.